Diversified Bottleneck Stock Research Report — V2

10 research candidates across 10 different themes

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Important: This report is for research and education only. It is not financial advice, not a recommendation to buy or sell securities, and does not consider personal risk tolerance, time horizon, tax situation, or portfolio. Small-cap and thematic stocks can be extremely volatile.

Research-queue note: researchPriorityTier (deprioritize | watch | prioritize | top_research | immediate_deep_dive), bottleneckResearchScore and the Jev route (pass | watch | reject) order research attention only. They are not buy, sell, strong-buy, timing or position-size signals, and this report contains no price targets or entry prices.

Executive summary

Premise: start from a measurable demand wave (AI optical networks, AI-driven chip capacity, power generation, naval shipbuilding, nuclear fuel, electrification metals, magnets, defense autonomy, aero engines, medical sterilization), find the layer with the lowest qualified-output ceiling, and ask whether the selected company controls that layer and captures value. **Classification mix: 1 Potentially Hard to Scale (LITE), 9 Probably Temporary.** Lumentum (LITE, new this cycle) is the only name to clear all five mandatory gates, but four of its five gates cleared by less than 0.10, so it is a low-margin call. ASML failed only the qualified-supply-lag gate, at 0.74 against a 0.75 threshold, even though ASML said on 14 Oct that planned 2027 and 2028 supply will not fully meet demand: Jev weighted the company's own funded +30% capacity steps. BWXT failed only value capture (0.50). Treat results near the threshold as low-margin calls; run-to-run variation in Jev near 0.75 is expected.

Jev routing: 8 *pass* (active research list: LITE, ASML, BWXT, GEV, HWM, CCJ, MP, AVAV); 2 *watch* (secondary list: SHC, FCX); none *reject*, so no substitution was required. SHC and FCX were re-run once with added evidence (Q2 call transcript; Fastmarkets LME Week review) as the protocol requires; both stayed *watch*. **Dropped or not carried:** Vertiv (VRT, data-center power and cooling) was researched and run through Jev (Probably Temporary, route pass, score 59, up from 53 and watch on 4 Oct) but removed for hidden concentration because LITE takes the single AI data-center capex slot. Energy storage was not carried because the U.S.-listed integrator layer is showing execution problems rather than scarcity rents: Fluence cut its FY2026 revenue outlook to ~$2.4B (from $3.2–3.6B) and expects an adjusted EBITDA loss near $200M, citing its Houston thermal-management factory ramp and transformer and inverter delays (https://www.energy-storage.news/fluence-cuts-2026-revenue-ebitda-guidance-as-us-production-delays-bite/); its U.S. cells come from a privately owned Tennessee plant, so the cell layer is not reachable through a U.S. listing in this scan (inference). FLNC was a Jev *reject* on 21 Sep. Cybersecurity was not carried: no qualified-supply bottleneck candidate was identified (CRWD was a Jev *reject* on 21 Sep). **Principal uncertainty:** how fast funded capacity (InP fabs at Lumentum, Coherent and in Japan and China; EUV +30% steps; three-OEM turbine expansion; casting furnaces) closes gaps relative to demand that is itself concentrated in AI capex and government budgets.

One-theme-per-stock watchlist

Ranked by bottleneckResearchScore (computed in code from Jev Score features). Rank is a research-queue order only.

Rank	Ticker	Theme	Durability indicator	Confidence	Scanner thesis	Risk
1	**LITE**	Indium phosphide laser chips for AI optical interconnects	**Potentially Hard to Scale**	Medium	Lumentum makes indium phosphide (InP) laser chips that AI optical links need whether they use EMLs, silicon photonics or co-packaged optics; it reports demand exceeding qualified supply into 2028 and is raising prices, and Jev cleared all five gates, though four of them by less than 0.10.	High
2	**ASML**	EUV lithography systems for leading-edge logic and DRAM	**Probably Temporary**	Medium	ASML is the only commercial EUV supplier and on 14 Oct said planned 2027 and 2028 capacity 'will not fully meet' customer demand, while raising prices; Jev cleared four gates but put qualified supply lag at 0.74, one hundredth below the threshold, because ASML itself is funding large capacity steps.	Medium
3	**BWXT**	U.S. naval nuclear propulsion reactor components	**Probably Temporary**	Medium	BWXT is the established supplier of U.S. naval reactor components and the only maker of naval nuclear fuel, so time, supply, workaround and entry gates all cleared; but a single government customer limits pricing, and Jev put value capture at 0.50.	Medium
4	**GEV**	Heavy-duty gas turbine manufacturing slots for new power generation	**Probably Temporary**	Medium	Gas turbine commitments (116 GW) far exceed GE Vernova's 20–24 GW of annual output and Jev now clears the supply-lag gate, but every OEM and several smaller makers are expanding and BNEF warns of oversupply risk, so workaround lag (0.73) and value capture (0.67) failed.	Medium

Rank	Ticker	Theme	Durability indicator	Confidence	Scanner thesis	Risk
5	**HWM**	Superalloy turbine airfoil investment castings for jet engines	**Probably Temporary**	Medium	Howmet holds ~50% share in aerospace and gas-turbine blades and vanes and says it is at physical capacity limits, with strong capture, but it is adding the equivalent of ~2.5 plants and engine shortages are easing, so Jev failed supply lag (0.44) and workaround lag (0.61).	Medium
6	**CCJ**	Western uranium mining and conversion supply under long-term contracts	**Probably Temporary**	Medium	Term uranium prices are at the highest in Cameco's series and Kazakh supply is constrained, but uranium is a globally priced commodity where secondary supply, restarts and new projects respond, so Jev failed supply lag, workaround and value capture.	Medium-High
7	**MP**	Non-China rare earth separation and NdFeB magnet supply	**Probably Temporary**	Medium	MP is the scaled U.S. rare-earth producer moving into magnets, but ex-China light rare earths (NdPr) are oversupplied and below the government price floor, so economics depend on policy support and Jev failed value capture decisively (0.28).	Very High
8	**AVAV**	Loitering munitions and tactical autonomous strike systems	**Probably Temporary**	Medium	Army demand for loitering munitions is real (record $1.5B funded backlog), but DoD is deliberately buying cheap one-way attack drones from many vendors, so Jev failed all five gates; the route is now pass on stronger evidence, not stronger durability.	High
9	**SHC**	Cobalt-60 supply and contract sterilization capacity for medical devices	**Probably Temporary**	Medium	Sotera supplies Cobalt-60 and runs regulated sterilization capacity with >50% EBITDA margins, but management now calls Co-60 supply secure and says capacity is available in most places, which weakens the scarcity case; Jev routed SHC to watch twice.	High
10	**FCX**	Copper mine supply for electrification	**Probably Temporary**	High	Copper mine supply is under real strain (record-negative concentrate TCs, Chilean outages), but Freeport is a price-taking producer, ICSG forecasts widening refined surpluses and Grasberg returns through 2027, so Jev failed four gates and kept FCX on watch.	High

Durability cells always print the exact label; amber shading = Probably Temporary, teal = Potentially Hard to Scale. Color is never the only signal.

Jev research-queue metrics (research priority only — not investment advice)

Rank	Ticker	bottleneckResearchScore	researchPriorityTier	Route	Failed Noul gates (threshold 0.75)
1	LITE	87	immediate_deep_dive	pass	none
2	ASML	81	immediate_deep_dive	pass	supply lag 0.74
3	BWXT	77	top_research	pass	value capture 0.50
4	GEV	75	top_research	pass	workaround lag 0.73, value capture 0.67
5	HWM	74	top_research	pass	supply lag 0.44, workaround lag 0.61
6	CCJ	63	top_research	pass	supply lag 0.67, workaround lag 0.67, value capture 0.50
7	MP	59	top_research	pass	supply lag 0.70, workaround lag 0.68, value capture 0.28
8	AVAV	58	prioritize	pass	time gate 0.47, supply lag 0.61, workaround lag 0.46, entry barrier 0.37, value capture 0.41
9	SHC	54	prioritize	watch	supply lag 0.39, workaround lag 0.50, value capture 0.61
10	FCX	47	prioritize	watch	supply lag 0.40, workaround lag 0.35, entry barrier 0.35, value capture 0.21

Ranking uses unrounded scores; no ties this cycle.

V2 methodology

Demand versus qualified supply. Required bottleneck input = end demand × input intensity; qualified supply = nameplate × yield × uptime × customer-qualified share. Announced nameplate capacity is never treated as qualified output. **Three clocks.** A bottleneck is potentially durable only if the demand clock stays faster than both the qualified-supply clock (funding, build, yield, certification) and the workaround clock (redesign, substitution, recycling, second sourcing) over the stated horizon. **Seven tests** inform each judgment: multi-constraint, money-and-time, substitution-and-intensity, entry barrier, full cycle, bottleneck migration and value capture. **Industry scarcity is not company capture:** a hard industry problem with weak company capture is Probably Temporary at the stock level.

Conservative rule (code, not Jev). Each stock's evidence packet (dated source excerpts, URLs and source types) was sent to TypeSafe Jev (jev-1.13.0) for five Noul gates, six Score dimensions and a Choice route. Potentially Hard to Scale requires all five gate probabilities ≥ 0.75 (non-capital time gate, qualified supply lag, workaround lag, observed entry barrier, company value capture); anything else is Probably Temporary, and missing or mixed evidence counts as a failed gate. Code computes bottleneckResearchScore = 100 × (0.20·evidence/4 + 0.20·value capture/4 + 0.20·intensity/4 + 0.15·demand durability/4 + 0.15·exclusivity/4 + 0.10·research priority/4), +5 if Potentially Hard to Scale (cap 100), and ranks by it. Score never overrides a failed gate. Confidence and risk labels are analyst judgments under the framework.

Hidden-concentration check. Only one stock per primary spending pool. LITE is the single AI data-center capex name; VRT was dropped as the same pool (LITE kept on the higher code score, 87 versus 59). Disclosed partial overlaps: ASML's demand is AI-influenced but set by foundry and DRAM capex, a different buyer and constraint from data-center optics, though an AI capex slowdown would hit both; Howmet's gas-turbine sales (13% of revenue) ride the same power wave as GEV, but its primary driver is commercial aerospace (53%), and BNEF names single-crystal blades as an upstream turbine limit; BWXT (Navy propulsion), CCJ (uranium price) and SHC (reactor-made Co-60 for sterilization) draw on different budgets or prices; MP (rare earths) and FCX (copper) face different constraints.

Framework references (foundations, not stock evidence): Epoch AI, multiple constraints (https://epoch.ai/publications/can-ai-scaling-continue-through-2030); Howard Marks, high rewards attract supply (https://www.oaktreecapital.com/insights/memo/more-on-repealing-the-laws-of-economics); Mauboussin and Callahan, entry, scale and switching costs (https://www.morganstanley.com/im/publication/insights/articles/article_measuringthemoat.pdf).

Per-stock deep dives

Bracketed numbers such as [2] cite the numbered entries in that stock's Sources list. 'Inference' marks analyst reasoning; 'management claim' marks unverified company statements.

LITE — Lumentum Holdings Inc.

Theme: Indium phosphide laser chips for AI optical interconnects

Durability indicator: **Potentially Hard to Scale**

Confidence: Medium

Research horizon: 2–4 years **Risk:** High

Thesis

Lumentum makes indium phosphide (InP) laser chips that AI optical links need whether they use EMLs, silicon photonics or co-packaged optics; it reports demand exceeding qualified supply into 2028 and is raising prices, and Jev cleared all five gates, though four of them by less than 0.10.

Three-clock assessment

Demand: AI networks are moving to 800G/1.6T and later to co-packaged optics, all needing InP lasers [4]. Fiscal Q4 revenue was $1,006M, +109% YoY [1]; FQ1 FY27 guidance is $1.225–1.275B [2].

Qualified-supply: New capacity takes three to five years per the CEO [3]; Tokyo-area output rose ~12x in two years and the Greensboro InP conversion reaches first revenue only in early 2028 [2][3]. TrendForce says large-diameter InP substrate yield gains take years and analysts expect the imbalance beyond 2027 [4].

Workaround: Silicon photonics with CW lasers is gaining share at 1.6T but still needs an InP laser source, which Lumentum also sells [2][4]. Chinese InP fabs are expanding (Yuanjie, Accelink, San'an) but management sees no impact yet [2][4] (management claim).

Why the indicator may be right

- CEO: unable to meet ~70% of demand for some products into next year, with shortfalls for others through 2028; constraints worse than six months ago [3] (management claim, reported by Bloomberg).
- Pump lasers 'effectively sold out for the foreseeable future' with 70–80% share and mostly three-year, often take-or-pay LTAs [2] (management claim).
- Nvidia invested $2B each in Lumentum and Coherent, and hyperscalers are sharing capex, rather than qualifying new entrants [3][4].
- Jev gates: time 0.90, supply lag 0.77, workaround 0.83, entry 0.84, value capture 0.79 — all at or above 0.75.

Evidence against the indicator

- Coherent (also Nvidia-funded) and Japanese suppliers are expanding 6-inch InP capacity; one doubled wafer capacity and another targets 2028 mass production [4].
- Lumentum's own capacity is rising fast (12x in two years, Greensboro from 2028), which could close the gap if AI optics orders pause [2][3].
- Demand is concentrated in AI data-center capex and a few hyperscalers; China holds ~70% of refined indium and has tightened export controls [4].
- Four of the five gates (supply lag 0.77, value capture 0.79, workaround 0.83, entry 0.84) cleared by less than 0.10; demand durability scored only 2.54 of 4.

Company value capture

Strong to Mixed. Non-GAAP gross margin crossed 50% (50.4%) on mix, utilization and 'increased pricing on select products' [1][2]; FQ1 non-GAAP operating margin guided to 39.5–40.5% [2]. Offsets: a GAAP net loss of $7.2B from a non-cash convertible charge and strong buyer concentration [2].

What would flip the indicator

Coherent's and Japanese 6-inch InP additions or Chinese InP fabs qualifying at hyperscalers in 2027, or a pause in 1.6T/CPO orders, would push supply lag or value capture below 0.75 and flip the label to Probably Temporary. Hyperscaler share gains for laser-free or non-InP optical architectures would also flip it.

What to monitor

- EML and CW-laser lead times and price changes in FQ1 FY27 results
- Coherent and Japanese InP wafer capacity announcements
- Qualification of Chinese InP lasers at U.S. hyperscalers

Key risks (overall risk: High)

- AI capex cyclicality and hyperscaler concentration
- Competitor InP capacity (Coherent, Japan, China)
- Indium and InP export controls
- Valuation and expectation risk after +109% YoY growth (inference)

Jev second opinion

Model **jev-1.13.0** | durability (code): **Potentially Hard to Scale** | route: **pass** | bottleneckResearchScore: **87** | researchPriorityTier: **immediate_deep_dive** (research-queue label only, not investment advice). All five Noul gates cleared 0.75.

Noul gate	P(true)	Result
1. Non-capital time gate	0.90	PASS
2. Qualified supply lag	0.77	PASS
3. Workaround lag	0.83	PASS
4. Observed entry barrier	0.84	PASS
5. Company value capture	0.79	PASS

Score (0–4)	Value
evidence_strength	3.27
value_capture_strength	3.29
bottleneck_intensity	3.96
demand_durability	2.54
constraint_exclusivity	2.72
research_priority	3.86

Sources

[1] **Lumentum FQ4 and FY2026 results press release** — 2026-08-11 (primary): https://investor.lumentum.com/financial-news-releases/news-details/2026/Lumentum-Announces-Fourth-Quarter-and-Full-Fiscal-Year-2026-Results/default.aspx

[2] **Lumentum FQ4 2026 earnings call transcript (Investing.com)** — 2026-08-12 (primary): https://in.investing.com/news/transcripts/earnings-call-transcript-lumentum-tops-q4-2026-forecasts-as-ai-demand-lifts-outlook-93CH-5550417

[3] **Quartz (citing Bloomberg) — Lumentum sold out through early 2029** — 2026-10-09 (newswire): https://qz.com/lumentum-ai-data-center-optical-parts-sold-out-2029-100926

[4] **TrendForce — InP shortage emerges as AI optical interconnect bottleneck** — 2026-08-06 (independent): https://www.trendforce.com/news/2026/08/06/news-inp-shortage-emerges-as-ai-optical-interconnect-bottleneck/

ASML — ASML Holding N.V. (ADR)

Theme: EUV lithography systems for leading-edge logic and DRAM

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 3–5 years **Risk:** Medium

Thesis

ASML is the only commercial EUV supplier and on 14 Oct said planned 2027 and 2028 capacity 'will not fully meet' customer demand, while raising prices; Jev cleared four gates but put qualified supply lag at 0.74, one hundredth below the threshold, because ASML itself is funding large capacity steps.

Three-clock assessment

Demand: Leading-edge customers expect demand for their own products to outpace supply through at least 2028; some want EUV slots for 2029–2030 [1][2].

Qualified-supply: ~65 Low-NA EUV systems in 2026 versus 44 in 2025; capacity +30% for 2027 and more than +30% under study for 2028 [1][2]. Management says Zeiss optics is not the limit [2] (management claim).

Workaround: No alternate EUV supplier: UBS puts China's EUV effort near ASML's 2004 level and sees no equivalent within ~10 years [4]. A Chinese immersion DUV machine was reported in July, with analysts doubting its ramp [3].

Why the indicator may be right

- ASML is funding a large response (+30% for 2027, more under study for 2028), so the qualified-supply clock is speeding up [1][2].
- Demand is concentrated in one AI-driven Logic/DRAM wave; China (~20% of 2026 sales) faces export-control limits [2][3].
- Q3 net sales were slightly below LSEG consensus (EUR 11.66B versus 11.72B) [3].
- Jev qualified supply lag 0.74, just under 0.75; the other four gates cleared.

Evidence against the indicator

- The company says planned 2027 and 2028 supply 'will not fully meet the demand of our customers' [1].
- Price increases on 'the most critical products' were completed, with some effect in 2027 and more in 2028 [2].
- No competing EUV source within ~10 years per UBS [4]; entry-barrier gate 0.91, value capture 0.86.

Company value capture

Strong. Q3 gross margin 57.5%, net income EUR 4.96B on net sales of EUR 11.66B [1]; negotiated price increases [2]; installed-base sales EUR 3.29B [1].

What would flip the indicator

Evidence that the 2028 expansion beyond +30% is not achievable, or customers booking 2029–2030 slots at higher prices, would lift supply lag above 0.75 and make it Potentially Hard to Scale. AI capex cuts or a credible non-ASML EUV path would deepen the Temporary reading.

What to monitor

- Q4 2026 bookings and 2028 capacity decision
- Pricing effect disclosed with 2027 guidance (January)
- China export-control changes and Chinese lithography progress

Key risks (overall risk: Medium)

- Semiconductor capex cyclicality
- Export controls and China exposure
- Execution of the capacity ramp (Q4 margin dip on ramp costs) [2]

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **81** | researchPriorityTier: **immediate_deep_dive** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): supply lag.

Noul gate	P(true)	Result
1. Non-capital time gate	0.80	PASS
2. Qualified supply lag	0.74	FAIL
3. Workaround lag	0.84	PASS
4. Observed entry barrier	0.91	PASS
5. Company value capture	0.86	PASS

Score (0–4)	Value
evidence_strength	2.92
value_capture_strength	3.04
bottleneck_intensity	3.72
demand_durability	2.21
constraint_exclusivity	4.00
research_priority	3.63

Sources

[1] **ASML Q3 2026 results press release (SEC Ex-99.1)** — 2026-10-14 (primary):
https://www.sec.gov/Archives/edgar/data/937966/000162828026065813/pressreleasefinancialresul.htm

[2] **ASML Q3 2026 earnings call transcript (Motley Fool)** — 2026-10-14 (primary):
https://www.fool.com/earnings/call-transcripts/2026/10/14/asml-asml-q3-2026-earnings-call-transcript/

[3] **CNBC — ASML says it's getting order interest through 2030** — 2026-10-14 (newswire):
https://www.cnbc.com/2026/10/14/asml-3q-earnings-ai-demand-orders-capacity.html

[4] **Asia Business Daily — UBS: China's EUV technology at ASML's 2004 level** — 2026-09-03 (independent):
https://www.asiae.co.kr/en/article/2026090309382047396

BWXT — BWX Technologies, Inc.

Theme: U.S. naval nuclear propulsion reactor components

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 5+ years **Risk:** Medium

Thesis

BWXT is the established supplier of U.S. naval reactor components and the only maker of naval nuclear fuel, so time, supply, workaround and entry gates all cleared; but a single government customer limits pricing, and Jev put value capture at 0.50.

Three-clock assessment

Demand: Navy targets one Columbia and two Virginia-class submarines a year [5]; BWXT won a $1.285B long-lead procurement (first of five annual task orders through 2030) and a $189M naval fuel contract on 1 Oct [3][4].

Qualified-supply: GAO says the submarine industrial base still cannot support the build-rate goal despite more than $10B of DOD investment, with late materials and quality problems [5].

Workaround: No practical substitute for naval nuclear propulsion; BWXT's Erwin, TN facility makes the fuel for all U.S. naval reactors [4].

Why the indicator may be right

- Monopsony buyer: contract structure and Navy budgets, not scarcity, set economics (inference).
- Shipyard delays can defer BWXT work; backlog was modestly lower sequentially on award timing [2][5].
- Contract risks include modification, termination and funding [4].
- Jev value capture 0.50; the other four gates cleared (0.80–0.90).

Evidence against the indicator

- Backlog $8.4B (+40% YoY), trailing book-to-bill 1.7x [1][2].
- BWXT has delivered more than 420 naval reactor cores and its NFS plant is a Category 1 NRC-licensed facility [3][4].
- No observed new entrant; GAO describes heavy sole-source reliance [5].

Company value capture

Mixed. Raised 2026 guidance (adjusted EBITDA $662–672M, non-GAAP EPS $4.70–4.80, free cash flow $345–360M) [1][2], but government contracting likely caps scarcity rents (inference).

What would flip the indicator

Evidence that Navy contract economics improve with scarcity (rising Government Operations margins on new awards, or customer-funded capacity on favorable terms) would be needed to lift the value-capture gate.

What to monitor

- FY2027 Navy shipbuilding appropriations
- Government Operations margin and backlog conversion in Q3 results
- GAO/Navy updates on Columbia and Virginia schedules

Key risks (overall risk: Medium)

- Single-customer and federal budget risk
- Shipyard schedule slips
- Nuclear quality and execution
- Competition in commercial nuclear lines

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **77** | researchPriorityTier: **top_research** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): value capture.

Noul gate	P(true)	Result
1. Non-capital time gate	0.90	PASS
2. Qualified supply lag	0.80	PASS
3. Workaround lag	0.87	PASS
4. Observed entry barrier	0.90	PASS
5. Company value capture	0.50	FAIL

Score (0–4)	Value
evidence_strength	2.62
value_capture_strength	2.44
bottleneck_intensity	3.86
demand_durability	2.65
constraint_exclusivity	3.94
research_priority	3.01

Sources

[1] **BWXT Q2 2026 results exhibit 99.1 (SEC)** — 2026-08-03 (primary): https://www.sec.gov/Archives/edgar/data/1486957/000148695726000041/bwxt_63026xerexhibit991.htm

[2] **BWXT Q2 2026 earnings call transcript** — 2026-08-03 (primary): https://stockanalysis.com/stocks/bwxt/transcripts/660315-q2-2026/

[3] **BWXT — $1.4B contracts for Naval Nuclear Propulsion Program** — 2026-05-07 (primary): https://investors.bwxt.com/news-releases/news-release-details/bwxt-announces-14-billion-contracts-naval-nuclear-propulsion

[4] **BWXT — awarded $189 million naval nuclear fuel contract** — 2026-10-01 (primary): https://www.bwxt.com/bwxt-awarded-189-million-naval-nuclear-fuel-contract/

[5] **GAO-26-109068 Navy and Coast Guard Shipbuilding** — 2026-04-22 (independent): https://files.gao.gov/reports/GAO-26-109068/index.html

GEV — GE Vernova Inc.

Theme: Heavy-duty gas turbine manufacturing slots for new power generation

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 3–5 years **Risk:** Medium

Thesis

Gas turbine commitments (116 GW) far exceed GE Vernova's 20–24 GW of annual output and Jev now clears the supply-lag gate, but every OEM and several smaller makers are expanding and BNEF warns of oversupply risk, so workaround lag (0.73) and value capture (0.67) failed.

Three-clock assessment

Demand: Orders $24.2B (+88% organic); gas backlog and slot reservations rose from 100 to 116 GW, at least 125 GW expected by year-end [1].

Qualified-supply: 20 GW per year from Q3 2026, 24 GW in 2028, actions toward 30 GW in 2030 [1]; Siemens Energy lead times three years or more [3]; Mitsubishi deliveries 2028–2030 [4].

Workaround: BNEF tracks 44 plants across 26 makers; announced expansions lift output more than 50% by 2030, and non-big-three makers supply ~20% of additions [5].

Why the indicator may be right

- Announced global expansions add ~35 GW of annual capacity by 2030 [5].
- BNEF: 'Today's scarcity does not rule out oversupply tomorrow'; past booms ended in plant closures and bankruptcies [5].
- Slot reservations are not firm orders until converted [1][2].
- Jev workaround lag 0.73 and value capture 0.67 failed; supply lag 0.80 passed.

Evidence against the indicator

- Contracted gigawatts are roughly five times annual output (116 GW versus 20–24 GW) [1].
- Peers are sold out into 2028–2030 and report improving profitability on new orders [3][4].
- Power EBITDA margin +240 bp to 18.8% on higher volume and price [1]; BNEF says the limits are now upstream (blades, forgings, alloys, labor) [5].

Company value capture

Mixed. Pricing and large slot-reservation down payments drove $5.1B of Q2 free cash flow [1][2], but bidding among three expanding OEMs is competitive.

What would flip the indicator

Price increases on new slots plus high reservation-to-order conversion through 2027 would support the workaround and capture gates. Falling lead times or order cancellations would deepen the Temporary reading.

What to monitor

- Q3 2026 results (reservation conversion, pricing)
- Peer and second-tier capacity announcements
- Data-center load forecasts and behind-the-meter alternatives

Key risks (overall risk: Medium)

- Data-center load forecasts disappointing
- OEM capacity race compressing pricing
- Execution on the output ramp
- Upstream part shortages (blades, forgings) [5]

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **75** | researchPriorityTier: **top_research** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): workaround lag, value capture.

Noul gate	P(true)	Result
1. Non-capital time gate	0.87	PASS
2. Qualified supply lag	0.80	PASS
3. Workaround lag	0.73	FAIL
4. Observed entry barrier	0.82	PASS
5. Company value capture	0.67	FAIL

Score (0–4)	Value
evidence_strength	3.11
value_capture_strength	3.08
bottleneck_intensity	3.72
demand_durability	2.52
constraint_exclusivity	1.94
research_priority	3.29

Sources

[1] **GE Vernova 2Q 2026 results press release (SEC exhibit)** — 2026-07-22 (primary):
https://www.sec.gov/Archives/edgar/data/1996810/000199681026000147/gevpressrelease2q26.htm

[2] **GE Vernova 2Q 2026 earnings call transcript** — 2026-07-22 (primary):
https://www.gevernova.com/sites/default/files/gev_webcast_transcript_07222026.pdf

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HWM — Howmet Aerospace Inc.

Theme: Superalloy turbine airfoil investment castings for jet engines

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 3–5 years **Risk:** Medium

Thesis

Howmet holds ~50% share in aerospace and gas-turbine blades and vanes and says it is at physical capacity limits, with strong capture, but it is adding the equivalent of ~2.5 plants and engine shortages are easing, so Jev failed supply lag (0.44) and workaround lag (0.61).

Three-clock assessment

Demand: Commercial aerospace is 53% of revenue (+28% YoY) and gas turbines 13% (+38%) [1][2]; management calls gas-turbine demand 'extraordinary' [1].

Qualified-supply: Building several of the world's largest casting furnaces, ~2.5 plants of added capacity, ~30% yield gains year to date [3]; a ~$70M Texas expansion was reported 13 Oct [6].

Workaround: GE Aerospace is buying CPP for $11.75B to secure casting capacity [4]; Safran says LEAP engine shortages are over for 2026–27 [5].

Why the indicator may be right

- Rapid capacity and yield gains are themselves a fast supply response [3][6].
- Safran: 'the times of engine shortages are over' [5].
- OEM vertical integration (GE–CPP) adds captive supply and bargaining power [4].
- Jev failed supply lag (0.44) and workaround lag (0.61); value capture 0.80 passed.

Evidence against the indicator

- CEO: operating at physical capacity limits; a fifth blade-and-vane supplier is 'probably very unlikely' [3] (management claim).
- Long-term contracts with all major turbine and aero-engine OEMs through the decade [3].
- BNEF lists single-crystal blades and vanes among the upstream limits on turbine delivery [7]; Engine Products margin 37.7% (+470 bp) [1].

Company value capture

Strong. Engine Products adjusted EBITDA $517M (+51%) at a 37.7% margin [1]; long-term contracts define price and share [3].

What would flip the indicator

Capacity sold out beyond 2028 despite the new furnaces and CPP integration would lift the supply-lag gate. Margin compression or OEM insourcing would deepen the Temporary reading.

What to monitor

- Engine Products margin and capex in Q3 results
- GE–CPP closing and integration
- LEAP/GTF delivery rates and gas-turbine blade lead times

Key risks (overall risk: Medium)

- OEM concentration and vertical integration
- Aerospace cycle
- Gas-turbine exposure shared with GEV's power wave
- Rising capex needs (capex to increase in 2027) [1]

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **74** | researchPriorityTier: **top_research** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): supply lag, workaround lag.

Noul gate	P(true)	Result
1. Non-capital time gate	0.87	PASS
2. Qualified supply lag	0.44	FAIL
3. Workaround lag	0.61	FAIL
4. Observed entry barrier	0.86	PASS
5. Company value capture	0.80	PASS

Score (0–4)	Value
evidence_strength	2.48
value_capture_strength	3.41
bottleneck_intensity	3.17
demand_durability	2.39
constraint_exclusivity	3.09
research_priority	3.28

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CCJ — Cameco Corporation

Theme: Western uranium mining and conversion supply under long-term contracts

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 5+ years **Risk:** Medium-High

Thesis

Term uranium prices are at the highest in Cameco's series and Kazakh supply is constrained, but uranium is a globally priced commodity where secondary supply, restarts and new projects respond, so Jev failed supply lag, workaround and value capture.

Three-clock assessment

Demand: UxC long-term contracting ~33 Mlb in H1 2026 versus ~27 Mlb a year earlier [2]; Cameco says utilities are not at replacement rate [3]. A non-binding U.S.-Korea framework contemplates up to eight large U.S. reactors [4].

Qualified-supply: Kazatomprom is negotiating 2027 sulphuric acid supply under new Russian export approvals [7]; Jefferies cites financing, capex inflation and delays at development projects [6].

Workaround: Secondary supply and restarts respond to price; new U.S. ISR production is ramping (inference).

Why the indicator may be right

- Spot (~$89.6/lb) sits below term (~$96.5/lb) and 2026 contracting has lagged norms [5][6].
- Kazatomprom expects no material 2026 production impact from the acid measures [7].
- The U.S.-Korea reactor framework is non-binding and subject to definitive agreements, sites and financing [4].
- Jev failed supply lag (0.67), workaround lag (0.67) and value capture (0.50).

Evidence against the indicator

- Long-term price US$96.50/lb on 30 Sep, the highest in the series [5].
- Market-related contracts carry floors in the high $70s and ceilings around $160 [3].
- Cameco's 49% Westinghouse stake would gain from AP1000 builds and fuel fabrication under the framework [4].

Company value capture

Mixed. Contract floors and ceilings and H1 uranium adjusted EBITDA of $676M versus $641M [1][3], but no product differentiation in a commodity market.

What would flip the indicator

Replacement-rate contracting alongside failed project ramps and Kazakh constraints into 2027 could lift the supply-lag gate. A Kazakh or secondary-supply surplus would deepen the Temporary reading.

What to monitor

- UxC term volumes versus consumption
- Kazatomprom 2027 acid supply and guidance
- Definitive agreements under the U.S.-Korea framework

Key risks (overall risk: Medium-High)

- Kazakhstan and Russia supply-chain geopolitics
- Commodity price cycle
- Reactor construction delays
- Mill and reagent disruptions

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **63** | researchPriorityTier: **top_research** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): supply lag, workaround lag, value capture.

Noul gate	P(true)	Result
1. Non-capital time gate	0.83	PASS
2. Qualified supply lag	0.67	FAIL
3. Workaround lag	0.67	FAIL
4. Observed entry barrier	0.78	PASS
5. Company value capture	0.50	FAIL

Score (0–4)	Value
evidence_strength	2.25
value_capture_strength	2.30
bottleneck_intensity	3.12
demand_durability	2.45
constraint_exclusivity	2.30
research_priority	2.77

Sources

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MP — MP Materials Corp.

Theme: Non-China rare earth separation and NdFeB magnet supply

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 5+ years **Risk:** Very High

Thesis

MP is the scaled U.S. rare-earth producer moving into magnets, but ex-China light rare earths (NdPr) are oversupplied and below the government price floor, so economics depend on policy support and Jev failed value capture decisively (0.28).

Three-clock assessment

Demand: NdPr sales 1,006 t (+127%) in Q2 [1]; GM is validating Independence magnets with commercial deliveries targeted at year-end [2][4].

Qualified-supply: ~1,000 t per quarter of NdPr with full run-rate targeted by year-end; heavy rare earth (Dy/Tb) output targeted by year-end; the government is 100% offtaker of the 10X plant [4].

Workaround: China says it will keep approving rare earth licenses to Europe and the U.S.-China truce was extended to January [5]; ex-China PrNd fell more than 13% in August [3].

Why the indicator may be right

- Ex-China PrNd prices slipped below the $110/kg floor in MP's government agreement [3].
- PPA income ($17.6M) is more than half of Materials adjusted EBITDA ($32.5M); net loss $(20.3)M [1].
- Commercial magnet deliveries have not yet started [2][4].
- Jev value capture 0.28; supply lag 0.70 and workaround lag 0.68 also failed.

Evidence against the indicator

- Ex-China heavy rare earths are scarce (dysprosium CIF Europe ~16x China) [3].
- CFO: announced Western magnet capacity would need ex-China NdPr output to roughly double [4] (management claim).
- GM calls automotive-grade magnets difficult to make reliably at scale [2]; time gate 0.84 and entry barrier 0.81 passed.

Company value capture

Weak to Mixed. Economics lean on the government price floor and offtake; Magnetics adjusted EBITDA was $7.5M [1].

What would flip the indicator

Commercial magnet sales at market prices above the floor plus heavy rare earth sales at ex-China premiums would be needed. A broader China export thaw or removal of policy support would deepen the Temporary reading.

What to monitor

- GM magnet shipments in Q4 2026
- Ex-China PrNd price versus the $110/kg floor
- U.S.-China truce status in January 2027

Key risks (overall risk: Very High)

- Policy and appropriations dependence
- Execution on Independence and 10X
- Rare earth price volatility
- China export-policy shifts [5]

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **59** | researchPriorityTier: **top_research** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): supply lag, workaround lag, value capture.

Noul gate	P(true)	Result
1. Non-capital time gate	0.84	PASS
2. Qualified supply lag	0.70	FAIL
3. Workaround lag	0.68	FAIL
4. Observed entry barrier	0.81	PASS
5. Company value capture	0.28	FAIL

Score (0–4)	Value
evidence_strength	2.19
value_capture_strength	1.98
bottleneck_intensity	3.21
demand_durability	1.79
constraint_exclusivity	2.38
research_priority	2.51

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AVAV — AeroVironment, Inc.

Theme: Loitering munitions and tactical autonomous strike systems

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 3–5 years **Risk:** High

Thesis

Army demand for loitering munitions is real (record $1.5B funded backlog), but DoD is deliberately buying cheap one-way attack drones from many vendors, so Jev failed all five gates; the route is now pass on stronger evidence, not stronger durability.

Three-clock assessment

Demand: Funded backlog $1.5B (+37%), book-to-bill 1.4 [1]; Army plans ~$1.2B on LASSO over FY26–FY31 [5]; Switchblade 300 is being integrated on Abrams [6].

Qualified-supply: AV is expanding capacity with FY2027 capex at 12–14% of revenue and a $100M new campus [2].

Workaround: Drone Dominance aims for more than 200,000 UAS by 2027; Gauntlet 2 tested 23 platforms from 19 companies [4].

Why the indicator may be right

- Many vendors are clearing DoD evaluations for low-cost attack drones [4].
- The Army is looking to add more platforms to LASSO [5].
- GAAP net loss $(5.1)M in FQ1 [1].
- Jev failed all five gates (highest: supply lag 0.61).

Evidence against the indicator

- Switchblade 400 selected for LASSO and a $51M Switchblade 600 order [3][5].
- Switchblade 600 fielded across more than 20 brigade combat teams [3].
- Management reports rising international loitering-munition demand [2] (management claim).

Company value capture

Mixed. Gross margin 26% (21% a year earlier) and adjusted EBITDA $53.4M [1], but thin GAAP profitability and competitive procurement.

What would flip the indicator

Sole-source, multi-year production awards for LASSO or vehicle-mounted munitions that low-cost drones cannot meet would be needed. Share loss to Gauntlet winners would deepen the Temporary reading.

What to monitor

- LASSO production decisions
- Drone Dominance phase awards and unit prices
- AV gross margin and capacity ramp

Key risks (overall risk: High)

- Budget and continuing-resolution risk
- Disruption by low-cost drones
- Capacity execution
- Acquisition integration (BlueHalo)

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **pass** | bottleneckResearchScore: **58** | researchPriorityTier: **prioritize** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): time gate, supply lag, workaround lag, entry barrier, value capture.

Noul gate	P(true)	Result
1. Non-capital time gate	0.47	FAIL
2. Qualified supply lag	0.61	FAIL
3. Workaround lag	0.46	FAIL
4. Observed entry barrier	0.37	FAIL
5. Company value capture	0.41	FAIL

Score (0–4)	Value
evidence_strength	2.17
value_capture_strength	2.30
bottleneck_intensity	2.44
demand_durability	2.67
constraint_exclusivity	2.18
research_priority	2.27

Sources

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SHC — Sotera Health Company

Theme: Cobalt-60 supply and contract sterilization capacity for medical devices

Durability indicator: **Probably Temporary**

Confidence: Medium

Research horizon: 5+ years **Risk:** High

Thesis

Sotera supplies Cobalt-60 and runs regulated sterilization capacity with >50% EBITDA margins, but management now calls Co-60 supply secure and says capacity is available in most places, which weakens the scarcity case; Jev routed SHC to watch twice.

Three-clock assessment

Demand: Steady device and pharma sterilization demand; ~50% EO and ~30% gamma by single-use device [3]. Q2 revenue +9.2% and FY2026 outlook raised [1].

Qualified-supply: Co-60 cannot be stockpiled (half-life ~5.27 years) and new PWR sources take ~4.5 years [5]; but the CEO has 'a lot of confidence' in short-, mid- and long-term Co-60 supply [3]. The Salem PWR program was awaiting NRC authorization as of January [4].

Workaround: Sotera's own X-ray site is online with revenue from Q3 and another due in 2027–28; the main competitor is also adding X-ray [2][3].

Why the indicator may be right

- Management says Co-60 supply is secure and development capex is past its peak [3].
- Capacity is available in most places; only U.S. EO large chambers are tighter [2].
- X-ray capacity, including Sotera's own and a competitor's, can cap gamma pricing over time [2][3].
- Jev failed supply lag (0.39), workaround lag (0.50) and value capture (0.61).

Evidence against the indicator

- Co-60 decay and multi-year reactor cycles cannot be bought down; contracts historically repriced up after disruptions [5].
- Pricing +4.3% at Sterigenics and +3.1% at Nordion; a large customer is outsourcing in-house EO to Sterigenics [2].
- Entry-barrier gate 0.83: EO permitting is difficult and new entrants focus on X-ray, not Co-60 [3].

Company value capture

Mixed to Strong. Adjusted EBITDA margin 51.6% and Nordion segment margin 56.9% [1][2]. Offsets: $2.2B total debt (net leverage 3.0x) [1] and EO litigation (Georgia appeal; California trial January or April 2027) [2].

What would flip the indicator

Co-60 shortages or multi-year contracts at rising prices while X-ray fails to take gamma volume would support the supply-lag and capture gates. Large X-ray build-outs taking validated volume would deepen the Temporary reading.

What to monitor

- Nordion harvest schedule and Co-60 pricing
- EPA final action on the EtO sterilizer rule [6]
- EO litigation (Georgia appeal, California trial)

Key risks (overall risk: High)

- EO litigation and possible new claims
- Leverage ($2.2B debt) [1]
- EPA and nuclear regulation
- Co-60 supply disruption

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **watch** | bottleneckResearchScore: **54** | researchPriorityTier: **prioritize** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): supply lag, workaround lag, value capture. Re-run once with added evidence (watch route); second-run values shown.

Noul gate	P(true)	Result
1. Non-capital time gate	0.86	PASS
2. Qualified supply lag	0.39	FAIL
3. Workaround lag	0.50	FAIL
4. Observed entry barrier	0.83	PASS
5. Company value capture	0.61	FAIL

Score (0–4)	Value
evidence_strength	1.52
value_capture_strength	2.61
bottleneck_intensity	1.78
demand_durability	2.56
constraint_exclusivity	2.97
research_priority	1.64

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FCX — Freeport-McMoRan Inc.

Theme: Copper mine supply for electrification

Durability indicator: **Probably Temporary**

Confidence: High

Research horizon: 3–5 years **Risk:** High

Thesis

Copper mine supply is under real strain (record-negative concentrate TCs, Chilean outages), but Freeport is a price-taking producer, ICSG forecasts widening refined surpluses and Grasberg returns through 2027, so Jev failed four gates and kept FCX on watch.

Three-clock assessment

Demand: ICSG projects refined usage growth of ~1.5% in 2026 and ~2.5% in 2027 [3].

Qualified-supply: Global mine output fell ~1% YoY in Jan–Jul (Chile –7%) with Escondida, El Teniente and Centinela disruptions [5]; Grasberg at ~67% of pre-incident rates, 80% by mid-2027, near full by end-2027 [1][2]; grades down 40% since 1991 [4].

Workaround: ICSG: refined surplus 104 kt in 2026 and 368 kt in 2027 as smelters ramp [3]; Oyu Tolgoi and Peru added supply [5].

Why the indicator may be right

- ICSG forecasts widening refined surpluses [3].
- Freeport earns through price, not control of a chokepoint; realized price expected above $6.50/lb in Q3 [1].
- Grasberg's recovery adds supply through 2027; U.S. operations ran below plan in Q3 [1][2].
- Jev failed supply lag (0.40), workaround (0.35), entry (0.35) and value capture (0.21).

Evidence against the indicator

- Concentrate TC index fell to $(272.20)/t on 9 Oct, so smelters are paying miners for concentrate [5].
- Escondida suspension, El Teniente avalanche and a Centinela strike hit supply in Sep–Oct [5].
- LME copper hit a record $14,875/t intraday on 10 Sep [5]; brownfield capital intensity +65% since 2020 [4].

Company value capture

Weak. Earnings move with copper and gold prices [1]; there is no contractual pricing power over the constraint.

What would flip the indicator

Persistent refined deficits despite Grasberg's recovery and rising smelter output, with returns holding through mid-cycle prices; still unlikely to clear the capture gate without pricing power.

What to monitor

- Q3 results on 27 Oct 2026 and Grasberg ramp
- Chilean outages (Escondida, El Teniente) and TC levels
- ICSG's next forecast update

Key risks (overall risk: High)

- Indonesia operations and geopolitics
- Commodity price cycle
- Tariff policy
- Cost pressure (unit net cash costs ~5% above estimate in Q3) [1]

Jev second opinion

Model **jev-1.13.0** | durability (code): **Probably Temporary** | route: **watch** | bottleneckResearchScore: **47** | researchPriorityTier: **prioritize** (research-queue label only, not investment advice). Hard-to-Scale blocked by failed gate(s): supply lag, workaround lag, entry barrier, value capture. Re-run once with added evidence (watch route); second-run values shown.

Noul gate	P(true)	Result
1. Non-capital time gate	0.89	PASS
2. Qualified supply lag	0.40	FAIL
3. Workaround lag	0.35	FAIL
4. Observed entry barrier	0.35	FAIL
5. Company value capture	0.21	FAIL

Score (0–4)	Value
evidence_strength	2.17
value_capture_strength	1.88
bottleneck_intensity	2.39
demand_durability	1.25
constraint_exclusivity	1.62
research_priority	1.62

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How to use the watchlist

This is a research funnel, not a recommendation list. A Probably Temporary bottleneck can still produce a short windfall for the company that holds it, and a Potentially Hard to Scale bottleneck can still be a poor investment if valuation, balance sheet or governance fail. Suggested sequence: (1) read the primary filings and releases cited in each deep dive; (2) check competitors, customers and regulators for capacity additions, redesigns or second sourcing that would disconfirm the thesis; (3) form an independent valuation view, which this report does not provide; (4) write down invalidation conditions in advance, using the flip conditions above. bottleneckResearchScore, researchPriorityTier and the Jev route order research attention only and must never be used as position-size or timing signals. Watch-routed names (SHC, FCX) need better evidence before deeper work.

Changes versus the 4 Oct 2026 list

Added: LITE (InP laser chips for AI optics), Potentially Hard to Scale and ranked first (score 87). **Removed:** VRT, for hidden concentration with LITE (not for weak evidence). **Kept and re-researched with new sources:** ASML, BWXT, GEV, HWM, CCJ, MP, AVAV, SHC, FCX. Score moves (4 Oct to 18 Oct): ASML 79 to 81 (supply lag 0.67 to 0.74 after the Q3 statement that 2027–28 supply will not meet demand); BWXT 76 to 77; GEV 70 to 75 (supply lag now passes at 0.80; workaround 0.73 and capture 0.67 still fail); HWM 67 to 74; CCJ 60 to 63; MP 60 to 59; AVAV 56 to 58 (route watch to pass); SHC 71 to 54 (route pass to watch, after management described Co-60 supply as secure); FCX 44 to 47 (still watch). Classification mix moved from 0 / 10 to 1 Potentially Hard to Scale / 9 Probably Temporary.

Selected sources

Every URL cited in the body, and no others. Grouped by stock in report order, then the screening source cited for the dropped energy-storage theme, then framework references cited in the methodology. A URL cited by two stocks is listed once, under the first. Freshness: all current-condition claims rely on sources dated 18 Apr 2026 or later. Four stock sources are older than ~6 months and are used only for structural or regulatory context, each paired with a newer source: NEI Salem Co-60 program (2026-01-30; SHC), IEA copper (2026-03-02; FCX), NextBeam Co-60 outlook (2026-03-13; SHC) and the Federal Register EtO proposal (2026-03-17; SHC).

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